PROJECT AEGIS: SYSTEM ARCHITECTURE & TECHNICAL BRIEF

FINTECH SECURITY • CYBER DECEPTION
STATUS: 100% PRODUCTION VERIFIED

A Unified ML-Driven Fraud Detection and Deceptive Honeypot Defence Platform for UPI Payments

UPES Major Project • Team SafeSight

The Problem: Conventional financial anti-fraud engines only score transactions *post-authorization*. This architectural flaw fails against **social engineering / remote coercion** (AnyDesk / TeamViewer screen sharing), **VPA typo-squatting** (paytmm@upi mimicking brands), **impossible travel card velocities**, and **automated credential stuffing bots** where victims willingly authorize malicious MPINs.

The Aegis Solution: Aegis introduces proactive **Pre-PIN contextual interception** and a synthetic **Deceptive Ghost Sandbox Honeypot**. Rather than alerting or outright blocking bots (which triggers attackers to rotate proxies), Aegis transparently routes automated threats into a cryptographic proof-of-work tarpit, stalling attackers while extracting forensic network intelligence.

1. MULTI-TIER PRE-PIN DEFENSE & DECEPTIVE HONEYPOT PIPELINE

TIER 1: APPROVE (Risk P < 0.30)	TIER 2: DYNAMIC FRICTION (0.30 ≤ P ≤ 0.70)	TIER 3: GATEWAY BLOCK (Risk P > 0.70)	TIER 4: DECEPTIVE HONEYPOT (Automated Bot Attack)
Direct Execution: Zero-friction payment path. Native MPIN clearance in **<15ms**. Verified genuine human interaction with legitimate merchants.	**Psychological Coercion Break:** Injects mandatory 5s cooldown modal, active phone call alert, payee authenticity verification, and acoustic warning.	**Upstream Interception:** Transaction termination before MPIN exposure. Halts AnyDesk screen-sharing fraud, unverified gallery QRs, and typo-squatting.	**Ghost Sandbox Decoy:** Zero 403 blocks. Traps bots in 2,400ms PoW tarpit, synthetic wire transfer exfiltration, 3/3 retry loop, and WebRTC STUN leak.

2. CORE ALGORITHMIC ENGINES & MATHEMATICAL FORMULATIONS

Typo-Squatting Wagner-Fischer Matrix (src/levenshtein_engine.py)
Dynamic programming matrix calculating edit distance: *D[i,j] = min(D[i-1,j]+1, D[i,j-1]+1, D[i-1,j-1]+δ)* against verified NPCI handle tokens. Flags brand similarity ≥ **0.75**.

Social Engineering Collect-Trap Filter (src/inference_engine.py)
Monitors inbound collect requests, amount ratios (*amt / 2000*), and sender spam reputation. Instantly flags high-risk reverse-payment manipulation schemes.

Behavioral Biometrics Sensor (static/js/bot-biometrics-engine.js)
Continuous 60Hz telemetry tracking mouse trajectory coordinates and flight time (*T_f*). Calculates directional **Shannon Entropy**: *H(D) = -∑ P(d_i) log&sub2; P(d_i)*. Detects synthetic robotic automation (**H < 0.60**) and 0ms clipboard injection (**T_f ≤ 2.5ms**).

Contextual Risk Fusion Engine (src/inference_engine.py)
Fuses raw XGBoost gradient-boosted trees with runtime mobile/browser indicators: Screen Sharing Active (1) + Phone Call Active (1) → Forces severe threat tier (**P ≥ 0.992**). Breaks psychological manipulation before MPIN capture.

3. PRODUCTION XGBOOST MODELS & EMPIRICAL BENCHMARKS

Model Target	Model File & Algorithm	Dataset & Features	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Unified UPI Scam	xgboost_fraud_model.json XGBoost Booster (Tree Depth=6)	unified_transaction_risk.csv 5,000 synthetic UPI records (5 features)	**99.92%**	98.36%	100.0%	**99.17%**	**1.0000**
Login Telemetry	xgboost_login_honeypot_model.json XGBoost Booster (Max Depth=5)	honeypot_login_training.csv 2,000 login sessions (Keystroke & Entropy)	**99.85%**	99.60%	100.0%	**99.80%**	**0.9998**
Biometric Bot Sensor	Shannon Entropy & Flight Time Sensor Dual-Threshold Heuristic	bot_telemetry_micro_samples.csv 10 microsecond key flight & cursor traces	**100.0%**	100.0%	100.0%	**100.0%**	**1.0000**

4. ACTIVE HONEYPOT DECEPTION & AUTONOMOUS BOT QUARANTINE

Cryptographic PoW Tarpit: When a bot credential-stuffs admin_bot@crawler.net or exhibits low entropy (H < 0.60), Aegis delays response by **2,400ms** in a cryptographic challenge loop, collapsing crawler velocity by **98.7%** without throwing suspicious 403 errors.

Autonomous Retry Quarantine (3/3 Limit): The sandboxed bot automatically loops through 4 transfer cycles against mule VPA (mule_sweep01@okaxis). Injects synthetic NPCI delays: 1,250ms → 1,400ms → 1,100ms. Hard quarantine locks the channel on Attempt 4 (AML_ERR_QUARANTINE).

WebRTC STUN Telemetry Extraction: Silently bypasses datacenter proxies (VPN/Tor) to harvest real public IPs (103.21.14.88), geolocations, and mule bank accounts, appending live forensics to data/honeypot_session_forensics.csv for banking SOC alert feeds.

5. TECH STACK, REPOSITORY ARTIFACTS & OPERATIONAL READINESS

Core Stack: Python 3 • Flask Microservice (src/app.py with 9 REST endpoints) • XGBoost 2.0 • Scikit-Learn • ReportLab • Tailwind CSS • Three.js WebGL (Interactive 3D Isometric Architecture Diagram at /architecture-3d) • HTML5 Canvas Gauges.
Chrome Bot Extension: Manifest V3 extension (bot_extension/) & Userscript featuring automated Selenium headless emulation, robotic linear mouse sweeps, credential stuffing injection, and autonomous wire retry loops.
Datasets & Models: 6 telemetry/forensic CSVs in data/ • 3 pre-trained XGBoost Booster JSON models in models/ • 100% test coverage via tests/test_engines.py and test_bot_csv.py.